TEACHING-LEARNING PLAN

PGF-02-R39

Date: August 12 – November 5, 2026

1. Identification

Area of knowledge: Global Economics

Teacher's name: Nicolas Lopez

Grade: 11

Term: First Term, 2026–2027

Plan code and version: PGF-02-R39, Versión 08

Language: English

2. Learning Objective

Learning Objective

Analyze uncertainty in Colombian and global economic and financial settings by characterising continuous random variables and their PDFs, CDFs, and supports; calculating probabilities from uniform, triangular, piecewise-linear, circular-arc, and exponential models; and calculating and interpreting theoretical means and variances. Use simulation and samples to estimate probabilities and compare sample statistics with theoretical values. Apply Student's t and standard normal distributions to critical values, central areas, and tails, selecting the appropriate distribution according to the available population information. Integrate continuous probability modelling, simulation, sampling analysis, and, where appropriate, portfolio-risk measures in a real-data investigation, communicating justified methods, predictive probability statements, assumptions, limitations, and evidence-based conclusions.

Assessment Criteria

- **C1.** Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.
- **C2.** Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.
- **C3.** Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.
- **C4.** Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.
- **C5.** Calculates and interprets the mean and variance of continuous distributions using standard formulas and functions.
- **C6.** Estimates interval probabilities using relative frequencies from generated samples and compares them with theoretical probabilities.
- **C7.** Calculates sample means and variances for different sample sizes and compares them with the theoretical mean and variance.
- **C8.** Uses Student's t-distribution tables, degrees of freedom, and one-tailed or two-tailed areas to determine critical values, central probabilities, and tail probabilities or probability ranges.
- **C9.** Calculates and compares critical values and probability areas from the standard normal and Student's t-distributions, selecting and justifying the appropriate distribution according to whether the population standard deviation is known or estimated from the sample.
- **C10.** Integrates continuous probability modelling, simulation, sampling analysis, and portfolio-risk measures to investigate an economic or financial question, justifying the selected methods and communicating results, assumptions, limitations, and evidence-based conclusions.

3. Conceptual standards of the disciplinary knowledge that supports the narrative's development

1. A continuous random variable has an interval or union of intervals as its support. Its PDF f is nonnegative and has total area one; $P(a \leq X \leq b) = \int_a^b f(x)\,dx$, while $P(X = x) = 0$. Its CDF $F(x) = P(X \leq x)$ accumulates PDF area. A discrete variable instead assigns probability mass to countable values.
2. Uniform densities are constant. Triangular and other piecewise-linear densities are normalized and their interval probabilities can be obtained from rectangles, triangles, trapezoids, complements, and sums split at breakpoints. Density height is not itself probability.
3. For a PDF whose graph is a circular arc, normalization and interval probabilities use circle geometry. Relevant regions may be semicircles or circular segments, with segment area obtained as sector area minus triangle area; the support is linear even though the density boundary is an arc.
4. For independent events occurring at a stable rate λ, $T \sim \mathrm{Exp}(\lambda)$ has $f(t) = \lambda e^{-\lambda t}$ for $t \geq 0$, $F(t) = 1 - e^{-\lambda t}$, and $P(T > t) = e^{-\lambda t}$. Suitability depends on a reasonably constant rate, independence, and a continuing mechanism.
5. For a continuous X, $E(X) = \int x f(x)\,dx$ and $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$, where $E(X^2) = \int x^2 f(x)\,dx$. Standard distribution functions may provide equivalent formulas; both measures must be interpreted in context and variance carries squared units.
6. In a generated sample, the relative frequency of an interval is the count inside it divided by sample size. It estimates the theoretical interval probability; differences are expected because of sampling variability and should generally become more stable as sample size grows.
7. The sample mean $\bar{x} = n^{-1} \sum x_i$ and sample variance $s^2 = (n-1)^{-1} \sum (x_i - \bar{x})^2$ vary from sample to sample. Comparing them across sample sizes with $E(X)$ and $\mathrm{Var}(X)$ reveals convergence without claiming that every larger sample must be closer.
8. Student's t distribution is symmetric, depends on degrees of freedom, and has heavier tails than the standard normal. Tables require careful translation among one-tailed area, two-tailed area, central probability, and critical value.
9. Use the standard normal distribution when the population standard deviation is known and Student's t when it is estimated from the sample in the relevant inference setting. Compare critical values and probability areas on the same tail convention and justify the choice from the information given.
10. A defensible investigation begins with a focused question and obtainable real-world data, then selects relevant continuous models, simulation, sampling comparisons, and portfolio-risk measures where appropriate. Predictive probability claims must follow from the data and model, and the report must disclose sources, assumptions, uncertainty, limitations, and evidence-based conclusions.

4. Pre-lesson and Context: C1 Discovery Opening

Framework: A 55-minute inquiry cycle: notice and name a variable, distinguish discrete and continuous supports, match PMF/PDF/CDF representations, and revise explanations after whole-class critique. This activates prior knowledge before formal continuous modelling.

Grouping and resources: Pairs use context cards adapted from the current C1 practice activity, support-number-line sheets, PMF/PDF/CDF cards, colored pencils, and the compact solved C1 handout only after discussion for feedback. OPEN/Teams may distribute accessible copies.

Products: Each pair submits variable-and-support cards and one corrected misconception; each student completes an exit response distinguishing mass from density and interpreting a CDF value in context.

Vocabulary and command words: random variable, support, discrete, continuous, PMF, PDF, CDF, density, cumulative probability; identify, classify, distinguish, represent, justify, interpret.

Opening sequence

Students define variables from economic and operational contexts, mark countable or interval supports, and distinguish PMF point masses from PDF areas. They explain that a positive PDF height is not the probability of an exact continuous value and read $F(x)$ as an accumulated probability. The verified current C1 contexts may support classification, but distribution-family recognition remains subordinate to the precise C1 distinction and interpretation. **DUA support:** sentence frames, discrete-dot and continuous-line templates, color-coded mass/area/accumulation diagrams, and labeled axes. **Extension:** explain why a rounded measurement may be recorded discretely although the underlying model is continuous. **Transition to N1:** Ask how a nonnegative curve over a stated support becomes a valid probability model and how its area answers an economic question.

5. Experience: Three Pedagogical Actions

Pedagogical Action N1 – Foundations of Continuous Probability

Criteria: C1–C4.

Role in sequence: Develop the theoretical foundations for C1–C4 through teacher explanations, worked examples, guided exercises, contextualized problems, and frequent group collaboration. Progress from continuous-variable representations to geometric probabilities, circular-arc geometry, and exponential waiting-time models, then conclude with the First Learning Evidence assessing C1–C4.

Materials: Use the verified current criterion-focused C1, C2, C3, and C4 practice activities and the cumulative C1–C4 solved set where their content matches the revised criteria. The compact solved C1 handout is reinforcement only. Each practice activity remains focalized on its stated criterion; no C5 practice is included in N1. No unsolved C1–C4 First Learning Evidence exam is claimed.

Vocabulary: support, PDF, CDF, normalization, uniform, triangular, piecewise-linear, circular arc, circular segment, exponential rate, suitability.

C1 – Continuous variables, support, PDF, and CDF

Objective and theory: Distinguish continuous from discrete variables and interpret support, PDF, and CDF in economic contexts. For continuous X, state $f(x) \geq 0$, $\int f(x)\,dx = 1$, $P(a \leq X \leq b)$ as area, and $F(x) = P(X \leq x)$.

Procedure and practice: Name the variable and units; state its support; decide whether values are countable or measured on a continuum; identify the representation; interpret a PDF interval area and a CDF value in words. Use relevant current C1 classification contexts and selected cumulative prompts.

Errors, support, and evidence: Address classification by context rather than variable, missing support, and confusing height with point probability. Provide word banks, support diagrams, and mass-versus-area sketches. Evidence is an individual written explanation containing the variable, support, classification, representation, and contextual interpretation.

C2 – Uniform, triangular, constant, and piecewise-linear densities

Objective and theory: Normalize constant or piecewise-linear PDFs and calculate interval probabilities geometrically. Split at breakpoints and use rectangle, triangle, trapezoid, or complement areas.

Procedure and examples: Define variable/support, infer the shape, normalize total area, write the complete PDF including zero outside support, shade and split the requested interval, calculate its area, and interpret. The current C2 activity provides verified microcredit disbursement, coffee-auction premium, last-mile order volume, merchant reconciliation, and weekly payout-liquidity contexts.

Errors, support, and evidence: Address normalizing only the requested region, omitting breakpoints, and confusing heights with areas. Groups receive pre-drawn graphs and area cards before individual transfer problems. Evidence is a normalized piecewise PDF, labeled graph, geometric calculation, and contextual interpretation.

C3 – Circular-arc PDFs and circular segments

Objective and theory: Apply circular-arc density graphs to interval probability. Normalize the arc-bounded region and calculate requested subareas using circle, semicircle, sector, triangle, and circular-segment geometry.

Procedure and examples: Identify the circle's radius and center from the graph; relate PDF scale to total area one; mark interval endpoints; decompose the required region; use sector-minus-triangle when a circular segment is required; divide scaled subarea by the normalized total; interpret. Use the current C3 activity and the semicircular-arc items in the cumulative solved set only where they implement this geometry.

Errors, support, and evidence: Do not replace this criterion with circular means, resultant concentration, or merely periodic-variable work. Address confusing arc length with area and using degrees in radian formulas. Provide annotated circle diagrams and sector/triangle formula prompts. Evidence is a labeled decomposition, calculation, probability, and contextual statement.

C4 – Exponential waiting-time models

Objective and theory: Apply $T \sim \mathrm{Exp}(\lambda)$ to constant-rate waiting times using its density, CDF, survival function, and memoryless property, then assess suitability.

Procedure and examples: Convert rate units, identify the event as below/between/above, calculate with $F(t) = 1 - e^{-\lambda t}$ or $P(T > t) = e^{-\lambda t}$, interpret, and test the constant-rate and independence assumptions. Use the verified current C4 practice and matching cumulative tasks in service, logistics, and financial waiting-time contexts.

Errors, support, and evidence: Address rate/mean confusion, inconsistent units, incorrect complements, and uncritical model acceptance. Provide timelines and formula-choice prompts. Evidence includes calculations, units, interpretation, and an explicit suitability judgment.

N1 progression: 1. C1: distinguish discrete and continuous random variables and interpret support, PDF, and CDF. 2. C2: calculate probabilities for uniform, triangular, constant, and piecewise-linear densities using geometric areas. 3. C3: extend geometric probability to circular-arc densities and circular segments. 4. C4: model constant-rate waiting-time situations with the exponential distribution and evaluate appropriateness. Each step uses a practice activity focalized on that criterion before students proceed to the First Learning Evidence.

N1 assessment: The First Learning Evidence is an individual written exam assessing C1–C4 only. N1 instruction and practice end with C4; C5 begins in Pedagogical Action N2 and belongs to the Second Learning Evidence assessment group.

Pedagogical Action N2 – Theoretical Moments, Simulation, and Sampling

Criteria: C5–C7.

Role in sequence: Begin the Second Learning Evidence sequence with C5 theoretical mean and variance, then move to C6 simulation and C7 sampling. C5, C6, and C7 are taught and practised through separate criterion-focused activities before the individual C5–C7 exam. Students use and create interactive web tools as learning instruments where appropriate, explain the underlying reasoning, and work collaboratively with contextual economic or financial examples rather than treating automated output as a substitute for mathematics.

Materials and tool requirements: A separate criterion-focused C5 practice activity, C6 practice activity, and C7 practice activity must be prepared because no current standalone aligned activities are listed for these criteria. Interactive tools used for C6 or C7 may generate random samples from distributions studied in N1; allow parameters and sample sizes to change; estimate interval probabilities by relative frequency; compare simulated with theoretical probabilities; calculate sample means and variances; display their behavior as sample size changes; and compare them with theoretical mean and variance. Tool use does not merge the criteria: each activity remains focalized on the knowledge and skills of its stated criterion.

C5 – Theoretical mean and variance

Objective and theory: Calculate $E(X)$, $E(X^2)$, and $\mathrm{Var}(X)$ from a PDF or use the standard formulas for the named continuous distribution; interpret center and spread in context.

Procedure and practice: In a criterion-focused C5 activity, verify the PDF and support; select an integral or applicable distribution formula; calculate $E(X)$ and $E(X^2)$; obtain variance; check non-negativity and reasonable scale; and state units for the mean and squared units for variance. Previously studied uniform, triangular, piecewise-linear, circular-arc, and exponential models may provide contexts, but the activity target is C5 only.

Errors, support, and evidence: Address using $E(X^2)$ as variance, forgetting to square $E(X)$, and misreporting variance units. Provide formula organizers and partially completed integrals. Evidence is a transparent calculation and contextual interpretation of both measures. This standalone C5 practice activity must be prepared and is not claimed as an existing file.

C6 – Relative frequencies and theoretical probabilities

Objective and procedure: Before running the tool, students identify the distribution, parameters, support, interval, and theoretical probability. For several sample sizes they generate samples, count interval observations, calculate relative frequency, record the difference from theory, repeat trials, and explain sampling variation.

Practice, support, and evidence: In a criterion-focused C6 activity, groups test uniform, triangular, piecewise-linear, circular-arc, or exponential economic scenarios already understood theoretically, then each student analyzes a new run. Starter code or interface scaffolds, a count/n template, and side-by-side theoretical and empirical displays support access. Evidence includes settings, sample size, counts, hand-checked relative frequency, theoretical calculation, comparison, and interpretation. This standalone C6 practice activity must be prepared and is not claimed as an existing file.

C7 – Sample means and variances across sample sizes

Objective and procedure: Calculate $\bar{x}$ and unbiased s^2 for generated samples of different sizes, compare them with $E(X)$ and $\mathrm{Var}(X)$, repeat samples, and describe stability and sampling variability without asserting monotonic convergence.

Practice, support, and evidence: In a criterion-focused C7 activity, students inspect or write the tool's calculation logic rather than copying displayed values; groups predict effects of changing parameters and n, then verify and discuss. Provide formula templates, downloadable sample values, and plots of statistics against sample size. Evidence includes sample statistics for multiple sizes, at least one manual check, theoretical benchmarks, a comparison graph or organized list, and a reasoned conclusion. This standalone C7 practice activity must be prepared and is not claimed as an existing file.

N2 progression: 1. C5: complete a criterion-focused activity on theoretical mean and variance for continuous distributions. 2. C6: complete a criterion-focused activity on simulated relative frequencies compared with theoretical probabilities. 3. C7: complete a criterion-focused activity on sample means and variances for different sample sizes compared with theoretical values. These three focused stages lead to the Second Learning Evidence.

N2 assessment: The Second Learning Evidence is an individual written exam assessing C5–C7: C5 concerns theoretical mean and variance; C6 concerns simulated relative frequencies versus theoretical probabilities; and C7 concerns sample means and variances versus theoretical values. Students must show calculations and interpret supplied or generated sample output; possession of a functioning tool alone is not assessment evidence.

Material status: No current standalone criterion-focused C5, C6, or C7 practice activity, aligned C6/C7 web tool, or individual C5–C7 exam is listed in the Grade 11 Term 1 directory. Separate C5, C6, and C7 practice activities and the assessment materials must be prepared without relabeling obsolete standard-normal or inverse-normal legacy activities as current evidence.

Pedagogical Action N3 – Student's t, Normal Comparison, and Real-Data Investigation

Criteria: C8–C10.

Role in sequence: Use teacher explanations, worked examples, guided practice, and contextual situations to develop Student's t tables, degrees of freedom, tail conventions, critical values and probability areas, comparison with the standard normal, and justified distribution selection. The action culminates in a student-selected real-world investigation.

Materials: Teacher-prepared Student's t tables and aligned examples are required; the existing standalone normal table may support normal comparisons. No current C8–C9 exam or C10 brief is claimed. The legacy interactive-density/model-fitting project is not an assignable source for the revised C8–C10 evidence.

C8 – Student's t tables, degrees of freedom, and areas

Objective and theory: Use $df = n - 1$ where applicable and translate among one-tailed area α, two-tailed area α, central probability $1 - \alpha$, and positive/negative critical values. Recognize symmetry and heavier tails.

Procedure and practice: Sketch and shade the requested region; identify degrees of freedom; determine whether the table heading is one-tail, two-tail, or central area; read or bracket the critical value/probability; attach the correct sign or symmetric pair; interpret. Practice includes economic and financial mean contexts with varied df and tail wording.

Errors, support, and evidence: Address confusing n with df, halving the wrong area, and reading a one-tail column for a two-tail request. Use annotated table headers, area-conversion diagrams, and worked examples. Evidence is a shaded curve, df, table pathway, value or range, and interpretation.

C9 – Standard normal versus Student's t

Objective and theory: Calculate and compare critical values and areas for Z and t, then select the appropriate distribution: standard normal when the population standard deviation is known and Student's t when it is estimated from the sample in the stated inference setting.

Procedure and practice: Identify whether σ is known or estimated; standardize with the applicable standard error; match tail convention; calculate or table-read both distributions when comparison is requested; explain the effect of degrees of freedom; select and justify. Contextual guided problems use the same confidence or tail area so comparisons are meaningful.

Errors, support, and evidence: Address choosing by sample size alone, confusing s and σ, and comparing mismatched areas. Provide a known-σ/estimated-σ decision organizer and paired curve sketches. Evidence includes both values/areas where asked and a justification tied to the supplied population information.

C10 – Investigation selected in the C8–C9 exam

Phase 1 connection: The final thinking-and-writing question of the individual C8–C9 exam requires each student to propose a specific economic or financial situation and an investigable question using real-world obtainable data. The context selected in that exam is mandatory and cannot be replaced in Phase 2 by a predetermined project or generic coding assignment.

Phase 2 procedure: Research exclusively the selected topic; obtain and document appropriate real-world data; define variables and the probability question; use relevant term techniques to conduct a probability-based analysis; produce meaningful predictive probability statements supported by the data and appropriate modelling; and, where appropriate to the question, incorporate simulation, sampling analysis, continuous models, and portfolio-risk measures. Justify every method and communicate results, assumptions, limitations, and evidence-based conclusions.

Support and evidence: Support includes a data-source check, question-to-method conference, analysis organizer, prediction/evidence prompts, and draft feedback without changing the selected topic. Evidence is the sourced dataset, transparent calculations or tool output, model and method justification, predictive probability statements, limitations, and the final individual report. C10 is validated through this report.

N3 progression: C8 table conventions and t probabilities precede C9 comparisons and distribution selection. The individual exam assesses C8–C9 and fixes the topic; the subsequent report integrates appropriate term learning to validate C10.

6. Learning Evidences, Feedback, and Progression

First Learning Evidence — Individual C1–C4 written exam

Prior learning: Contextual examples, guided exercises, group problem solving, and individual transfer tasks cover C1–C4 through criterion-focused practice: C1 practice develops continuous-variable/PDF/CDF interpretation; C2 practice develops uniform, triangular, constant, and piecewise-linear geometric probability; C3 practice develops circular-arc and circular-segment probability; and C4 practice develops exponential calculation and suitability. Current C1–C4 files may supply these focused activities, while the solved cumulative set is never administered with solutions visible. C5 is not introduced or practised as part of N1.

Assessed product: An individual written exam assessing C1–C4 only: continuous-variable/PDF/CDF interpretation, uniform/triangular/piecewise geometric probability, circular-arc/segment probability, and exponential calculation and suitability.

Feedback and progression: Return criterion-coded comments; students classify errors as concept, representation, geometry, formula, units, calculation, interpretation, or suitability, then correct a parallel item and explain the revision. After completing and receiving feedback on the C1–C4 evidence, students begin N2 with the separate C5 practice activity. C5 theoretical values then become benchmarks for the separate C6 and C7 practice activities.

Second Learning Evidence — Individual C5–C7 written exam

Prior learning: Completion of the First Learning Evidence for C1–C4 is followed by N2 instruction and criterion-focused practice for C5, C6, and C7. C5 practice develops calculation and interpretation of theoretical mean and variance; C6 practice develops simulated relative frequencies versus theoretical interval probabilities; and C7 practice develops sample means and variances across sample sizes versus theoretical values. Interactive tools, prediction before simulation, manual checks, repeated runs, and parameter or sample-size comparisons support the relevant criterion without merging the practice activities.

Assessed product: An individual written exam assessing C5–C7: theoretical mean and variance for continuous distributions; simulated relative frequencies versus theoretical interval probabilities; and sample means and variances for different sample sizes versus theoretical values. Questions require visible reasoning from supplied or generated data, not automated answers alone.

Feedback and progression: Feedback targets the calculation and interpretation of $E(X)$, $E(X^2)$, and $\mathrm{Var}(X)$; mean and variance units; denominator choice; interval counting; sample-variance convention; use of theoretical values as benchmarks; and overclaims about convergence. Students submit corrected theoretical moments, sample calculations, and a revised sampling-variability explanation before using this reasoning in N3 and C10.

Third Learning Evidence — Two phases for C8–C10

Phase 1 – Individual C8–C9 exam

The exam assesses Student's t tables, degrees of freedom, one- and two-tailed areas, critical values, central and tail probabilities or ranges, standard-normal comparisons, and justified distribution selection. Its final thinking-and-writing question requires a specific real-world economic or financial question answerable with obtainable data. The student's selected context becomes the mandatory Phase 2 topic.

Phase 2 – Individual C10 report

The student researches only the exam-selected topic, obtains appropriate real-world data, and applies relevant term techniques. The report makes meaningful predictive probability statements supported by collected data and appropriate modelling; incorporates simulation, sampling analysis, continuous models, and portfolio-risk measures where appropriate; and justifies methods while communicating assumptions, limitations, results, and evidence-based conclusions. This report, not an unrelated web project, validates C10.

Feedback and progression: Phase 1 feedback corrects table conventions and model choice without replacing the chosen topic. Phase 2 feedback checks data provenance, alignment of question/data/method, probability reasoning, predictive meaning, assumptions, uncertainty, limitations, and claim-to-evidence links. Submitted evidence includes the exam, topic statement, source/data record, analysis, report, feedback, and revision.

References and Material Status

Verified current Term 1 sources

- **Course map:** README.md.
- **C1 practice with solutions:** 1-practice_activities/G11_T1_C1_practice_activity1.tex.
- **C2 practice with solutions:** 1-practice_activities/G11_T1_C2_practice_activity2.tex.
- **C3 practice with solutions:** 1-practice_activities/G11_T1_C3_practice_activity3.tex.
- **C4 practice with solutions:** 1-practice_activities/G11_T1_C4_practice_activity4.tex.
- **Cumulative C1–C4 assessment-style solved set:** 2-learning_evidences/G11_T1_C1C2C3C4_exam0_solution.tex.
- **Compact C1 solved reinforcement:** 4-extra_material/G11_T1_C1_practice_activity1_print1_solved.tex.
- **Standard-normal reference usable for C9 comparisons:** 4-extra_material/normal-table.tex.

PDF documents

1-practice_activities

- `1-practice_activities/G11_T1_C1_practice_activity1.pdf`.
- `1-practice_activities/G11_T1_C1_practice_activity1_print0.pdf`.
- `1-practice_activities/G11_T1_C1_practice_activity1_print1.pdf`.
- `1-practice_activities/G11_T1_C2_practice_activity2.pdf`.
- `1-practice_activities/G11_T1_C3_practice_activity3.pdf`.
- `1-practice_activities/G11_T1_C4_practice_activity4.pdf`.

2-learning_evidences

- `2-learning_evidences/G11_T1_C1C2C3C4_exam0_solution.pdf`.
- `2-learning_evidences/G11_T3_C1C7_midterm_solution.pdf`.

4-extra_material

- `4-extra_material/G11_T1_C1_practice_activity1_print1.pdf`.
- `4-extra_material/G11_T1_C1_practice_activity1_print1_solved.pdf`.
- `4-extra_material/normal-table.pdf`.

Term 1 root

- `PGF-02-R39_G11_Global_T1_2026-2027-plain.pdf`.
- `PGF-02-R39_G11_Global_T1_2026-2027.pdf`.

Legacy-material boundary

Files prefixed old_ in the current directory encode earlier criteria, including standard-normal-only extensions, inverse-normal work, and an interactive-density/model-fitting project. They are not current activities or learning evidences for this plan and must not be issued as C5–C10 materials. New aligned criterion-focused C5, C6, and C7 practice activities, later C8–C10 instruction, and the three individual assessments/report framework described above must be prepared without inventing unavailable files, dates, or institutional requirements.